

CloudWatch Agent – Advanced EC2 Monitoring

To Begin with the Lab

Summary of the Lab

In this lab, an Amazon EC2 instance was launched and configured to use the CloudWatch Agent for advanced monitoring. An IAM role with the required permissions was created and attached to the EC2 instance so that the Systems Manager (SSM) Agent and CloudWatch Agent could function correctly. The CloudWatch Agent was installed and configured directly from the Amazon CloudWatch console to collect additional system-level metrics such as memory and disk utilization. Finally, the configuration was verified by confirming that the **CWAgent** namespace appeared in Amazon CloudWatch and that advanced metrics were successfully collected.

Understanding CloudWatch Agent in Amazon CloudWatch

The **CloudWatch Agent** is an AWS software agent that runs on Amazon EC2 instances and on-premises servers to collect operating system metrics and log files that Amazon CloudWatch cannot collect by default. It exists because the default EC2 monitoring only provides infrastructure metrics such as CPU utilization, network traffic, and disk I/O, while memory usage, disk space, swap usage, and process information require an additional agent. This enables administrators to monitor the complete health of their servers from a single CloudWatch dashboard. The result is faster troubleshooting, better visibility, and proactive alerting for both cloud and on-premises workloads.

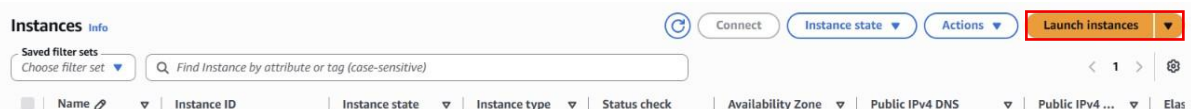
Example:

A production EC2 instance named **WebServer** hosts an e-commerce application where CPU utilization remains below 40%, but customers report slow response times. After installing the CloudWatch Agent, the administrator discovers that available memory has dropped below 10% and disk space on the application volume is nearly full. With these additional metrics, the root cause is identified quickly and resolved before the application becomes unavailable.

Lab Steps

Step 1: Launch an EC2 Instance

- Open the AWS Management Console.
- Go to **EC2**.
- Click **Launch Instance**.



- Enter the instance name (for example, **Web Server**).
- Select **Amazon Linux 2023 AMI**.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Web-server

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start



[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

- Choose **t2.micro** (or **t3.micro**) as the instance type.
- Select an existing key pair.
- Keep the default settings.
- Click **Launch Instance**.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

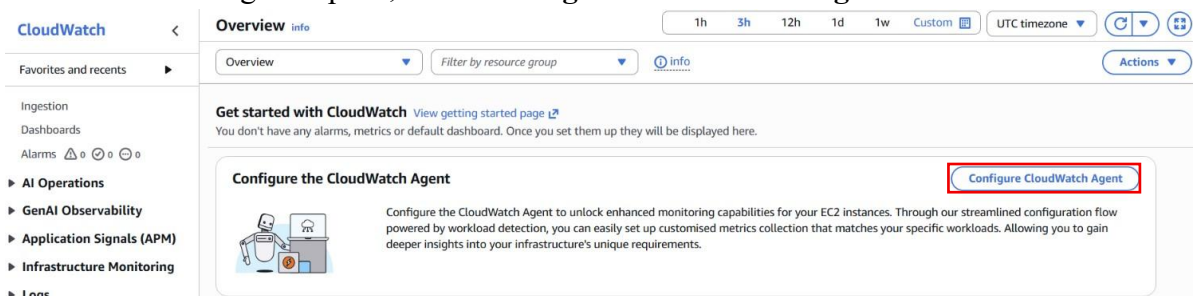
Key pair name - *required*

testkey

[Create new key pair](#)

Step 2: Open the CloudWatch Console

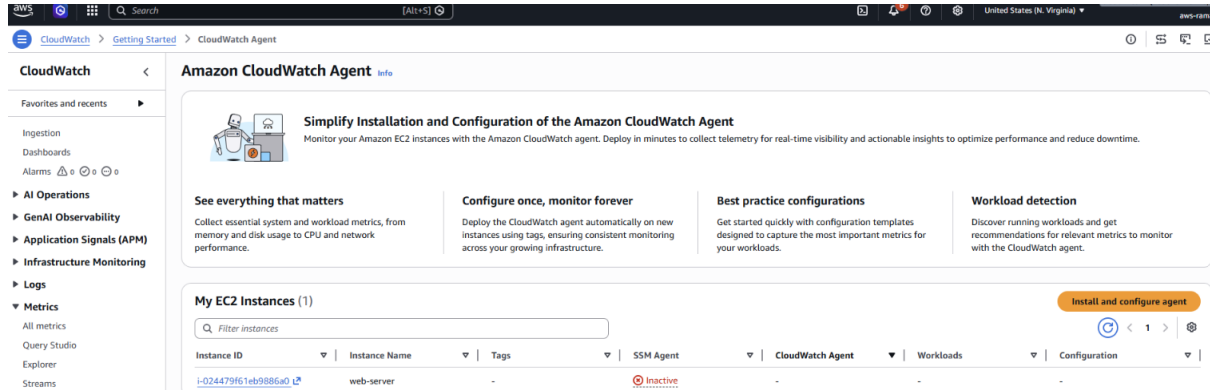
- Go to the **CloudWatch** service.
- In the left navigation pane, select **Configure CloudWatch Agent**.



- Wait until your EC2 instance appears in the list.

Step 3: Check the SSM Agent Status

- Locate your EC2 instance.
- Verify the **SSM Agent Status**.
- If the status is **Inactive**, an IAM role is required before CloudWatch Agent can be installed.

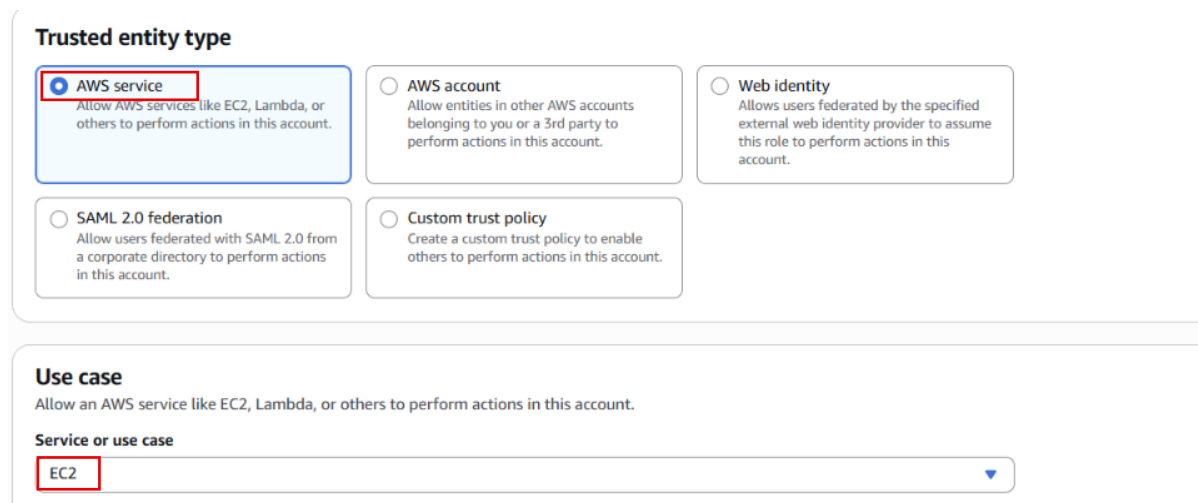


Step 4: Create an IAM Role

- Go to the **IAM** service.
- Select **Roles**.



- Click **Create Role**.
- Choose **AWS Service**.
- Select **EC2** as the trusted service.



- Click **Next**.

Step 5: Attach Required IAM Policies

- Search for and attach **AmazonSSMManagedInstanceCore**.
- Search for and attach **CloudWatchAgentServerPolicy**.

Permissions policy summary		
Policy name	Type	Attached as
AmazonSSMManagedInstanceCore	AWS managed	Permissions policy
CloudWatchAgentServerPolicy	AWS managed	Permissions policy

- Click **Next**.
- Enter the role name (for example, **CloudWatch-Agent-Role**).

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and "+,=,_,@,." characters.

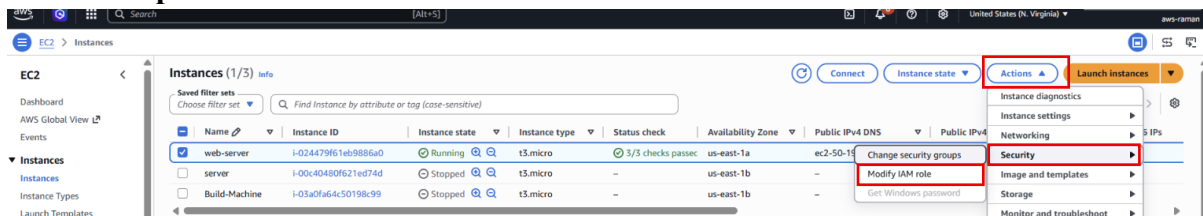
Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]\#%*^{}~`-''

- Click **Create Role**.

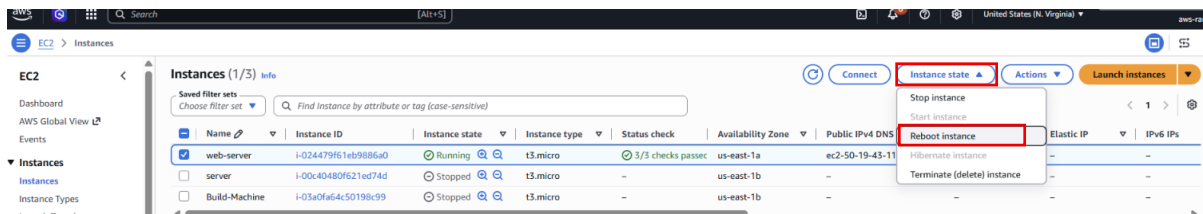
Step 6: Attach the IAM Role to the EC2 Instance

- Return to the **EC2** console.
- Select your EC2 instance.
- Choose **Actions** → **Security** → **Modify IAM Role**.
- Select **CloudWatch-Agent-Role**.
- Click **Update IAM Role**.



Step 7: Reboot the EC2 Instance

- Select the EC2 instance.
- Choose **Instance State** → **Reboot Instance**.



- Wait until the instance restarts.
- Return to the **CloudWatch** console.
- Confirm that the **SSM Agent Status** is now **Installed/Active**.

My EC2 Instances (1)							Install and configure agent
Filter instances							1
Instance ID	Instance Name	Tags	SSM Agent	CloudWatch Agent	Workloads	Configuration	
i-024479f61eb9886a0	web-server	-	Installed	Not installed	-	-	

Step 8: Install the CloudWatch Agent

- In CloudWatch, click **Install and Configure Agent**.

My EC2 Instances (1)							Install and configure agent
Filter instances							1
Instance ID	Instance Name	Tags	SSM Agent	CloudWatch Agent	Workloads	Configuration	
i-024479f61eb9886a0	web-server	-	Installed	Not installed	-	-	

- Select the EC2 instance.
- Click **Next**.
- Disable CloudWatchAgentServerPolicy
- Click **Install Agent**.

Configuration

Instance ID Instance Name Tags SSM Agent CloudWatch Agent Workloads

i-024479f61eb9886a0 web-server - Installed Not installed -

Installation settings

Enable telemetry publishing

Attach the CloudWatchAgentServerPolicy managed policy to your instances to grant the CloudWatch agent permissions to send telemetry to CloudWatch.

☐ Attach CloudWatchAgentServerPolicy to the selected EC2 instances

Cancel Install agent

- Wait until the installation completes successfully.

Selected Instances (1)							1
Instance ID	Instance Name	Tags	SSM Agent	CloudWatch Agent	Workloads	Configuration	
i-024479f61eb9886a0	web-server	-	Installed	Installed	EC2 (Linux)	Not configured	

Step 9: Configure the CloudWatch Agent

- Click **Configure CloudWatch Agent**.

CloudWatch

Get started

CloudWatch Agent

Install and configure

Step 1 Select instances

Step 2 Agent status

Step 3 Edit configuration

Step 4 Review and deploy

Agent status

Review the CloudWatch Agent installation status on your selected instances before proceeding to configuration.

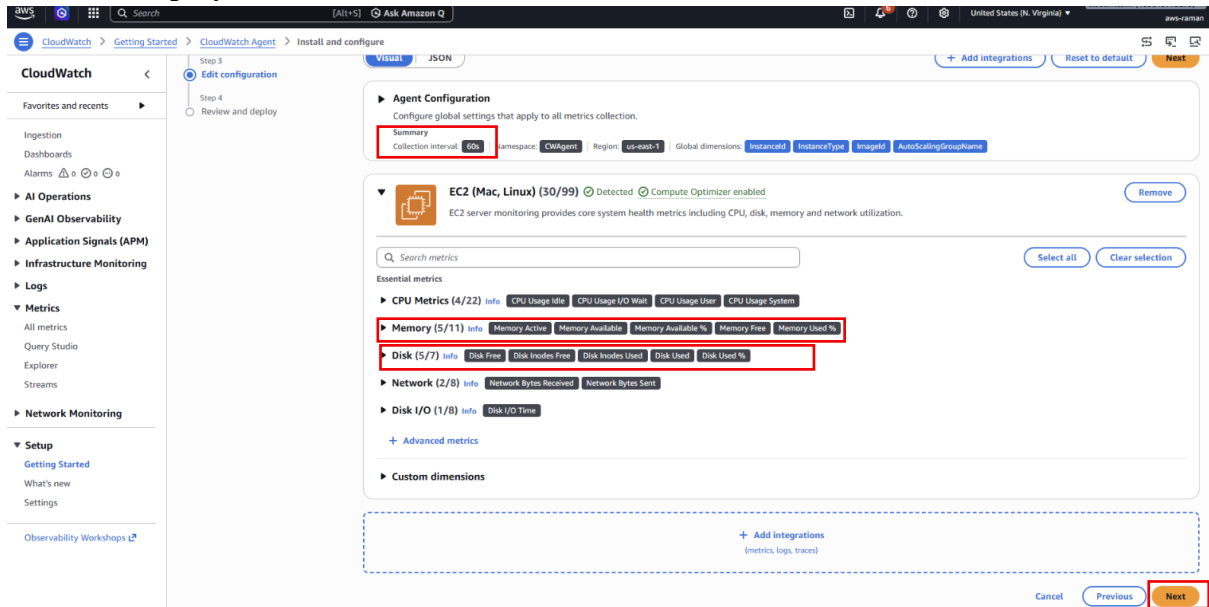
Selected Instances (1)

Instance ID	Instance Name	Tags	SSM Agent	CloudWatch Agent	Workloads	Configuration
i-024479f61eb9886a0	web-server	-	Installed	Installed	EC2 (Linux)	Not configured

Cancel Previous **Configure CloudWatch Agent**

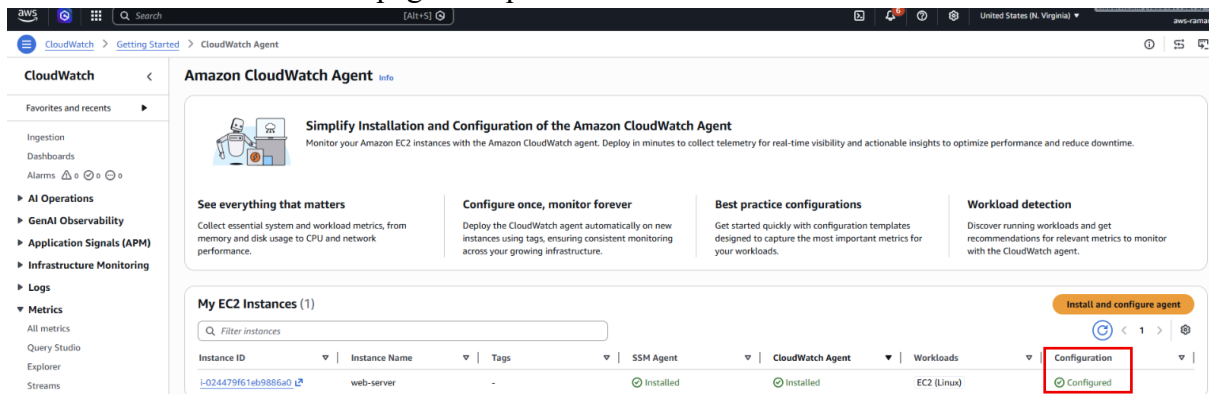
- Keep the default collection interval (**60 seconds**).
- Verify the metric namespace as **CWAgent**.
- Choose the metrics you want to collect, such as:
 - Memory Available
 - Memory Available Percentage

- 3.Memory Free
 - 4.Memory Active
 - 5.Disk Free
 - 6.Disk Usage
 - 7.Running Processes
- Click **Next**.
 - Click **Deploy**.



Step 10: Verify the Configuration

- Wait until the status changes to **Installed** and **Configured**.
- Refresh the CloudWatch page if required.



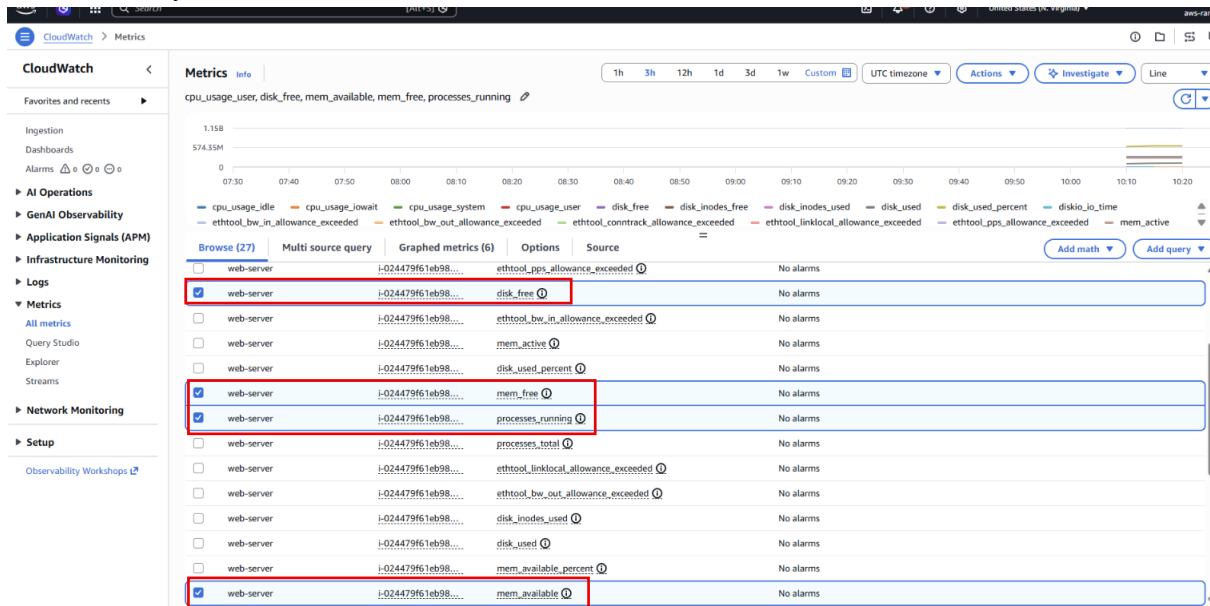
Step 11: View Advanced Metrics

- Go to **CloudWatch** → **Metrics** → **All Metrics**.
- Open the **CWAgent** namespace.
- Select your EC2 instance.
- View advanced metrics such as:
 - 1.Memory Usage
 - 2.Memory Available
 - 3.Memory Free
 - 4.Disk Space

5. Running Processes

6. CPU Time

7. Other system-level metrics



Step 12: Monitor the Instance

- Wait **10–20 minutes** for metrics to populate.
- Run applications or generate workload on the EC2 instance.
- Observe the changes in the CloudWatch graphs.

